



National Hospital & Medical Operations Management System (NHMOMS)

Language: ANSI C

Architecture: Structured Enterprise Console System

Storage: Binary File-Based Persistence

Version: 1.0



Executive Summary

The National Hospital & Medical Operations Management System (NHMOMS) is a structured healthcare enterprise simulation platform developed in ANSI C.

The system integrates hospital operational components including:

- Secure administrator authentication
- Patient record management
- Doctor management
- Appointment scheduling
- Pharmacy inventory control
- Billing generation
- Revenue analytics
- Timestamp-based audit logging

The platform models centralized hospital workflow using deterministic structured programming and binary file persistence without relying on external databases or frameworks.



System Architecture

NHMOMS follows a layered logical structure:

- Authentication Layer
- Patient Management Module
- Doctor Management Module
- Appointment Scheduling Module
- Pharmacy Inventory Module
- Billing Engine
- Revenue Analytics Engine
- Audit Logging Layer
- Binary Persistence Layer

Although implemented in a single source file, functional responsibilities are clearly separated.

Authentication Module

The system ensures secure administrative access.

Features:

- Automatic admin initialization on first execution
- Password hashing using djb2 algorithm
- No plaintext credential storage
- Binary verification against `hospital_users.dat`

Default Credentials:

Username: admin

Password: admin123

Hashing Logic:

```
hash = ((hash << 5) + hash) + character
```

Patient Management

Each patient record contains:

- Patient ID
- Name
- Age
- Gender
- Disease

Functional Capabilities:

- Add patient
- View patient records

Storage File: `patients.dat`

Doctor Management

Each doctor record contains:

- Doctor ID
- Name
- Specialization
- Consultation Fee

Functional Capabilities:

- Add doctor
- View doctor records

Storage File: `doctors.dat`



Appointment Scheduling

Appointment records include:

- Appointment ID
- Patient ID
- Doctor ID
- Date

Functional Capability:

- Create appointment

Storage File: `appointments.dat`

Note: Schedule conflict validation is not implemented in current version.



Pharmacy Inventory Management

Medicine records include:

- Medicine ID
- Medicine Name
- Quantity
- Price
- Reorder Level

Functional Capabilities:

- Add medicine
- View medicine inventory
- Low stock detection (quantity $\leq$ reorder level)

Storage File: `medicines.dat`

Low-stock condition displays visible warning.

Billing Engine

Billing records include:

- Bill ID
- Patient ID
- Treatment Cost
- Medicine Cost
- Total Bill
- Timestamp

Billing Formula:

$\text{Total Bill} = \text{Treatment Cost} + \text{Medicine Cost}$
--

Functional Capabilities:

- Generate bill
- Auto-calculate total
- Attach timestamp
- Persist record

Storage File: `bills.dat`

Revenue Analytics

The system performs statistical analysis on billing data.

Metrics Computed:

- Mean (Average Bill Value)
- Variance
- Standard Deviation

Formulas Used:

$\begin{aligned}\text{Mean} &= \Sigma \text{ Bills} / N \\ \text{Variance} &= \Sigma (\text{Bill} - \text{Mean})^2 / N \\ \text{Standard Deviation} &= \sqrt{\text{Variance}}\end{aligned}$

Purpose:

- Evaluate billing consistency
- Measure financial variability
- Assess revenue distribution



Audit Logging

All major activities are recorded in:

`hospital_logs.txt`

Logged Events:

- Admin login
- Patient addition
- Doctor addition
- Appointment creation
- Medicine addition
- Bill generation
- Revenue analytics execution
- System exit

Each log entry includes timestamp metadata.



Data Storage Design

Binary files used:

File	Purpose
hospital_users.dat	Authentication records
patients.dat	Patient records
doctors.dat	Doctor records
appointments.dat	Appointment records
medicines.dat	Pharmacy inventory
bills.dat	Billing records
hospital_logs.txt	Audit logs

Binary storage ensures:

- Structured record alignment
- Faster read/write performance
- Persistent data integrity
- Reduced storage overhead



Compilation & Execution

Compile:

```
gcc hospital_management_system.c -o hospital -lm
```

Run:

```
./hospital
```



Engineering Highlights

- ANSI C structured programming
- File-based database simulation
- Healthcare workflow abstraction
- Financial computation logic
- Statistical revenue analysis
- Deterministic memory usage
- Timestamp-based traceability



Project Significance

This system demonstrates the ability to design and implement a centralized hospital management workflow using low-level systems programming.

It reflects:

- Operational modeling capability
- Structured data persistence design
- Secure authentication handling
- Analytical computation integration
- Enterprise-level logic implementation



Conclusion

The National Hospital & Medical Operations Management System (NHMOMS) is a complete healthcare enterprise simulation platform implemented in pure ANSI C.

It integrates:

- Security
- Patient Management
- Doctor Management
- Appointment Scheduling
- Pharmacy Inventory
- Billing Engine
- Revenue Analytics
- Audit Logging

All within a structured, deterministic runtime architecture.